Thermodynamic description of the Al–Zn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Zn binary system as calculated with pycalphad from the SGTE dataset AlZn-93Mey-LB.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Al) + (Zn), lies at 380.8 °C.

1. Introduction

The Al–Zn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic, solid miscibility gap, monotectoid. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of S. an Mey, Z. Metallkd., 84, 451-55(1993). The calculations use pycalphad 0.11.2 at 1 atm.

For Al, the liquidus meets the axis at 660.3 °C. For Zn, the liquidus meets the axis at 419.5 °C.

2. Phase diagram

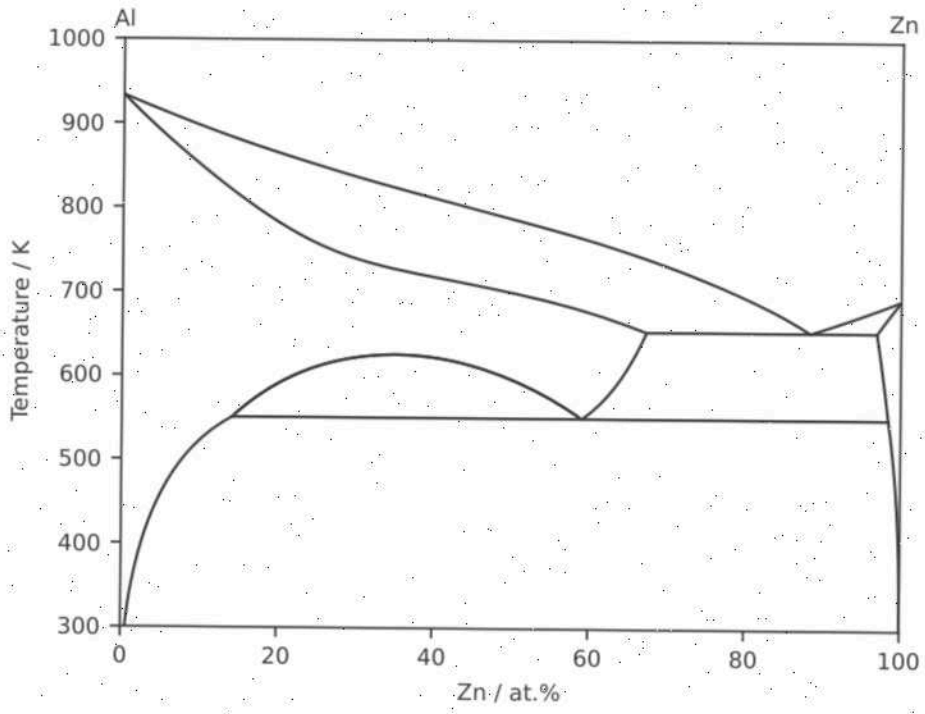


Fig. 1. Calculated Al–Zn phase diagram.

The dataset places a eutectic (L → (Al) + (Zn)) at 380.8 °C; the composition of L is x(Zn) = 0.884. We obtain 277.2 °C for the eutectoid reaction ((Al)2 → (Al)1 + (Zn)), where (Al)2 has x(Zn) = 0.590.

Table 1. Invariant reactions in the Al–Zn system (calculated).

Reaction	Type	T / °C	Compositions (at.% Zn)		
			phase 1	phase 2	phase 3
L → (Al) + (Zn)	eutectic	380.8	88.4	67.3	96.9
(Al)2 → (Al)1 + (Zn)	eutectoid	277.2	59.0	14.1	98.4

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 1000 K the integral mixing enthalpy of the liquid has its maximum, 2.62 kJ/mol, at x(Zn) = 0.500.

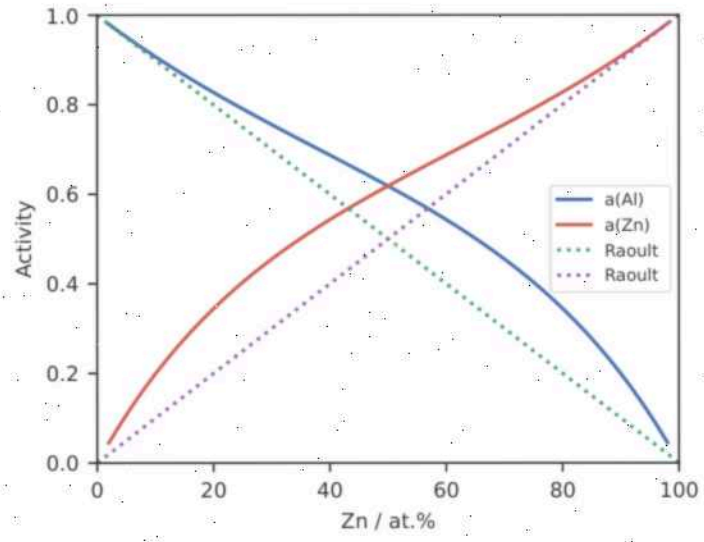


Fig. 2. Calculated activities in the liquid at 1000 K (reference: pure liquids).

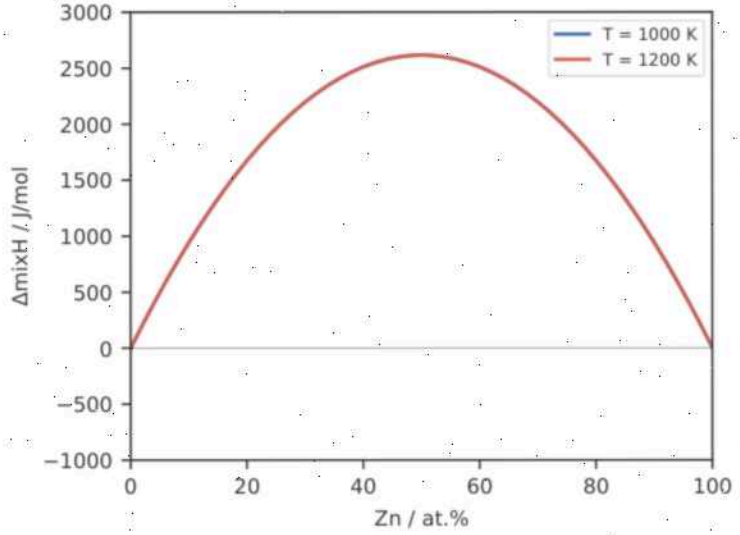


Fig. 3. Calculated mixing enthalpy of the liquid at 1000 K and 1200 K.

Table 2. Redlich–Kister parameters of the liquid (Al–Zn).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	a	b	c
L0	10465.55	−3.39259	0.0

Values as printed in the TDB file.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.